

LINUX BASICS — DEVOPS ASSIGNMENT

Student Details

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Subject: DevOps

Assignment: Linux Basics

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GitHub Repository: <https://github.com/KUSHRUP/Linux-Basics-DevOps.git>

Objective

The objective of this assignment is to practise essential Linux command-line operations, including file and directory management, viewing file contents, searching for patterns, compressing and extracting files, downloading files, changing file permissions, and working with environment variables.

All tasks were performed using the Linux terminal on Ubuntu.

Task 1 — Creating and Renaming Files/Directories

1. Create a directory named `test_dir`

Command

```
mkdir test_dir
```

Explanation

The `mkdir` command is used to create a new directory. The above command creates a directory named `test_dir` in the current working directory.

2. Create an empty file named `example.txt`

Command

```
touch test_dir/example.txt
```

Explanation

The `touch` command is used to create an empty file. Here, an empty file named `example.txt` was created inside the `test_dir` directory.

3. Rename `example.txt` to `renamed_example.txt`

Command

```
mv test_dir/example.txt test_dir/renamed_example.txt
```

Explanation

The `mv` command is used to move or rename files and directories. Here, `example.txt` was renamed to `renamed_example.txt`.

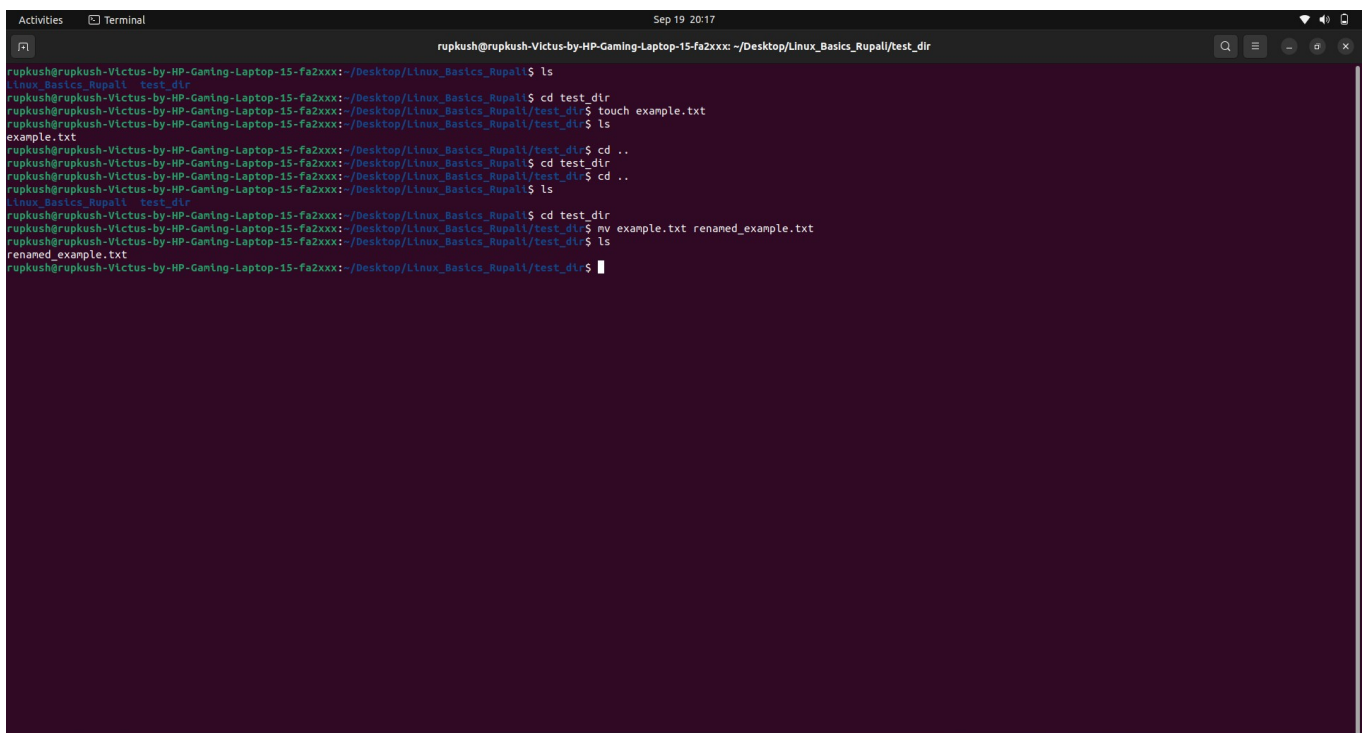
Verification

```
ls test_dir
```

Output

```
renamed_example.txt
```

Screenshot



```
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali/test_dir
Linux_Basics_Rupali test_dir
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$ cd test_dir
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali/test_dir$ touch example.txt
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali/test_dir$ ls
example.txt
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali/test_dir$ cd ..
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$ cd test_dir
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali/test_dir$ cd ..
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$ ls
Linux_Basics_Rupali test_dir
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$ cd test_dir
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali/test_dir$ mv example.txt renamed_example.txt
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali/test_dir$ ls
renamed_example.txt
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali/test_dir$
```

Task 2 — Viewing File Contents

4. Display the contents of /etc/passwd using cat

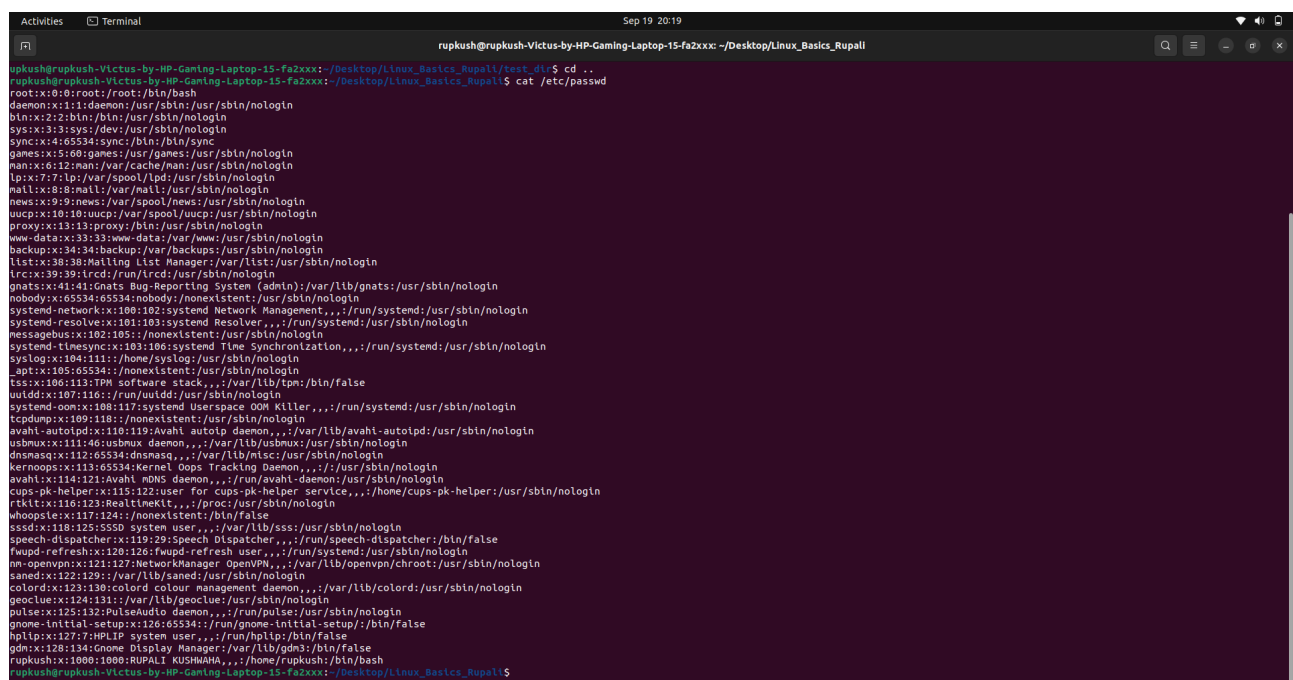
Command

```
cat /etc/passwd
```

Explanation

The `cat` command is used to display the contents of a file in the terminal. The `/etc/passwd` file contains information about user accounts configured on the Linux system.

Screenshot



```
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$ cd ..
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail list Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105:,:/run/systemd:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111:,:/home/syslog:/usr/sbin/nologin
dbus:x:105:65534:,:/run/dbus:/usr/sbin/nologin
tss:x:106:113:TPM software stack,,,:/var/lib/tpm:/bin/false
uidd:x:107:116:,:/run/uidd:/usr/sbin/nologin
systemd-oom:x:108:117:systemd Userspace OOM Killer,,,:/run/systemd:/usr/sbin/nologin
tcpdump:x:109:118:,:/usr/sbin/nologin
avahi-autoipd:x:110:119:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:111:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/mtsc:/usr/sbin/nologin
kernoops:x:113:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
avahi:x:114:121:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
cups-pk-helper:x:115:122:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
rtkt:x:116:123:Realtimekit,,,:/proc:/usr/sbin/nologin
whoopsie:x:117:124:,:/bin/false
sssd:x:118:125:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:119:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
fwupd-refresh:x:120:126:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
nm-openvpn:x:121:127:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
saned:x:122:129:,:/var/lib/saned:/usr/sbin/nologin
colord:x:123:130:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:124:131:,:/var/lib/geoclue:/usr/sbin/nologin
pulse:x:125:132:pulseaudio daemon,,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:126:65534:,:/run/gnome-initial-setup:/bin/false
hplip:x:127:7:HPLIP system user,,,:/run/hplip:/bin/false
gdm:x:128:134:Gnome Display Manager:/var/lib/gdm3:/bin/false
rupkush:x:1600:1600:RUPALI KUSHAWA,,,:/home/rupkush:/bin/bash
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$
```

5. Display the first 5 lines using head

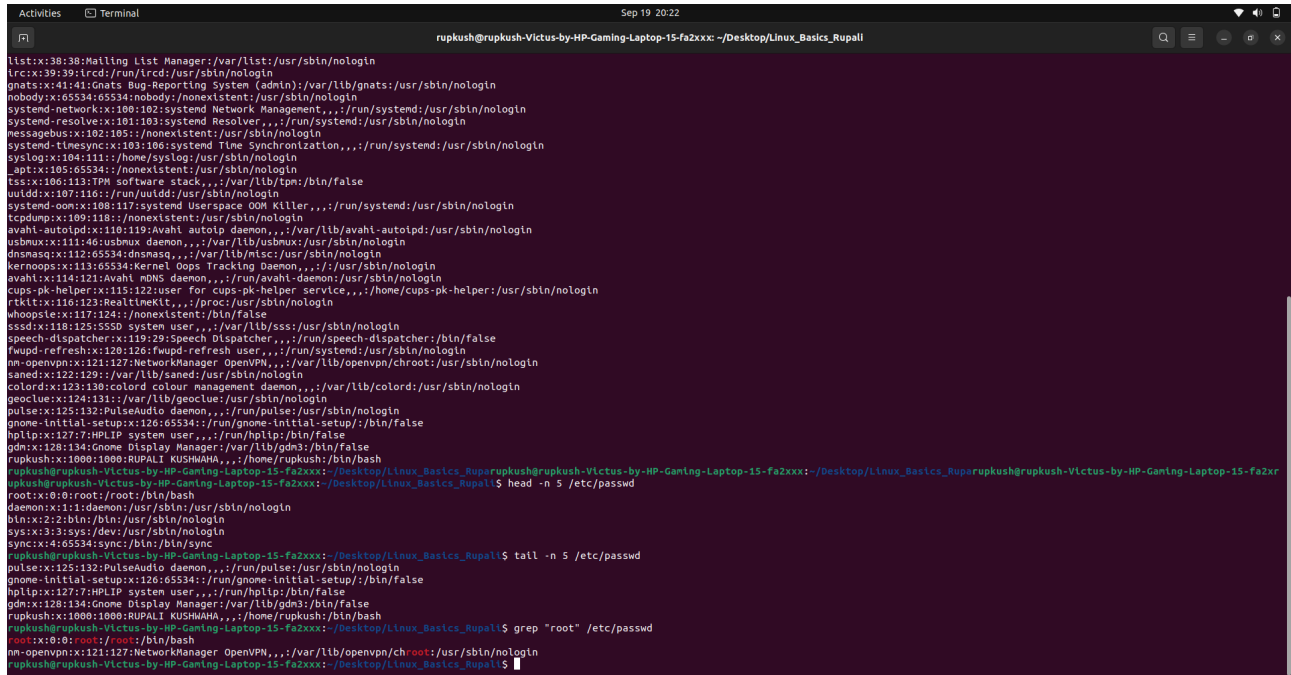
Command

```
head -n 5 /etc/passwd
```

Explanation

The `head` command displays the beginning of a file. The `-n 5` option specifies that only the first five lines should be displayed.

Screenshot



```
list:x:38:38:MailList Manager:/var/lib/mail:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
system-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
system-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:109:/:/nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111:/:/home/syslog:/usr/sbin/nologin
_apt:x:105:65534:/:/nonexistent:/usr/sbin/nologin
tss:x:106:113:TPM software stack,,:/var/lib/tpm:/bin/false
uiddd:x:107:116:/:/run/uiddd:/usr/sbin/nologin
systemd-oom:x:108:117:systemd Userspace OOM Killer,,,:/run/systemd:/usr/sbin/nologin
tcpdump:x:109:118:/:/nonexistent:/usr/sbin/nologin
avahi-autoipd:x:110:119:Avahi autoip daemon,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmuxd:x:111:46:usbmux daemon,,:/var/lib/usbmux:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,:/var/lib/misc:/usr/sbin/nologin
kernoops:x:113:65534:Kernel Oops Tracking Daemon,,:/usr/sbin/nologin
avahi:x:114:121:Avahi mDNS daemon,,:/run/avahi-daemon:/usr/sbin/nologin
cups-pk-helper:x:115:122:user for cups-pk-helper service,,:/home/cups-pk-helper:/usr/sbin/nologin
rtkit:x:116:123:RealtimeKit,,:/proc:/usr/sbin/nologin
whoopsie:x:117:124:/:/nonexistent:/bin/false
sssd:x:118:125:SSSD system user,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:119:29:Speech Dispatcher,,:/run/speech-dispatcher:/bin/false
fwupd-refresh:x:120:126:fwupd-refresh user,,:/run/systemd:/usr/sbin/nologin
nm-openvpn:x:121:127:NetworkManager OpenVPN,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
saned:x:122:129:/:/var/lib/saned:/usr/sbin/nologin
colord:x:123:130:colord colour management daemon,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:124:131:/:/var/lib/geoclue:/usr/sbin/nologin
pulse:x:125:132:PulseAudio daemon,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:126:65534:/:/run/gnome-initial-setup:/bin/false
hplip:x:127:7:HPLIP system user,,:/run/hplip:/bin/false
gdm:x:128:134:Gnome Display Manager:/var/lib/gdm3:/bin/false
rpkush:x:1000:1000:RUPALI KUSHWAHA,,:/home/rpkush:/bin/bash
rpkush@rpkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
rpkush@rpkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$ tail -n 5 /etc/passwd
pulse:x:125:132:PulseAudio daemon,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:126:65534:/:/run/gnome-initial-setup:/bin/false
hplip:x:127:7:HPLIP system user,,:/run/hplip:/bin/false
gdm:x:128:134:Gnome Display Manager:/var/lib/gdm3:/bin/false
rpkush:x:1000:1000:RUPALI KUSHWAHA,,:/home/rpkush:/bin/bash
rpkush@rpkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
nm-openvpn:x:121:127:NetworkManager OpenVPN,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
rpkush@rpkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$
```

6. Display the last 5 lines using tail

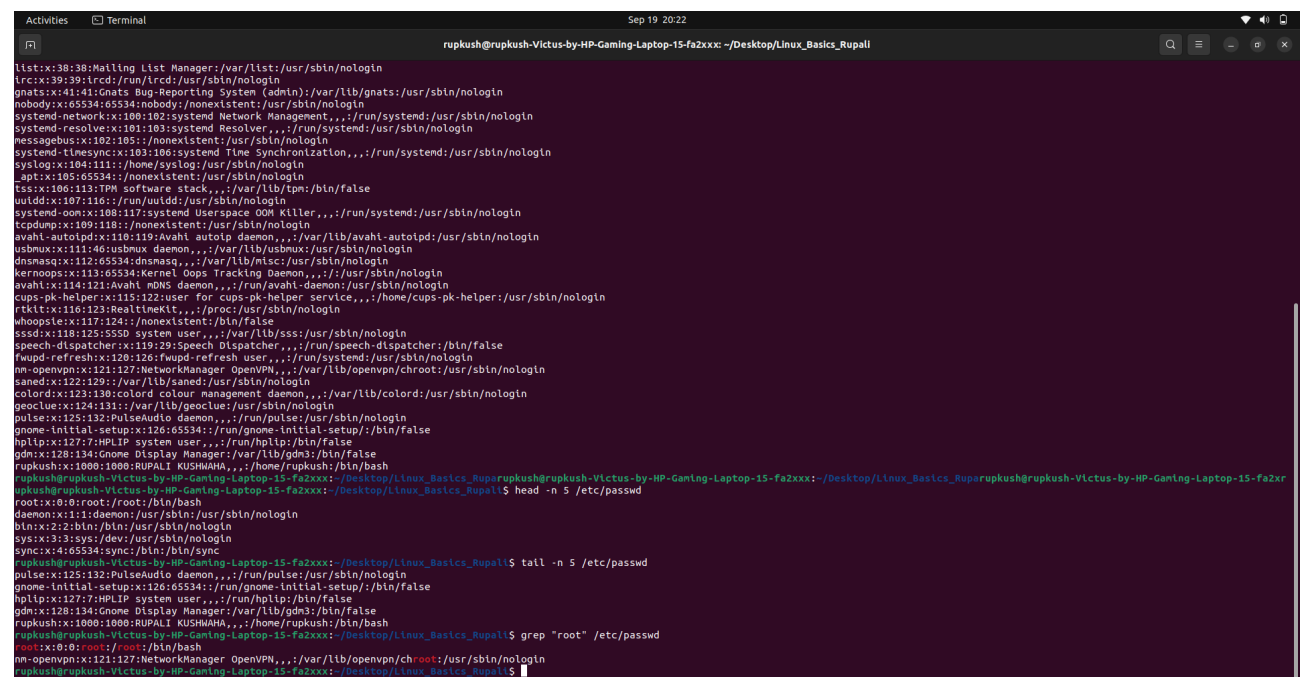
Command

```
tail -n 5 /etc/passwd
```

Explanation

The `tail` command displays the end of a file. The `-n 5` option specifies that only the last five lines should be displayed.

Screenshot



```
list:x:38:38:MailList Manager:/var/lib/mail:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
system-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
system-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:109:/:/nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111:/:/home/syslog:/usr/sbin/nologin
_apt:x:105:65534:/:/nonexistent:/usr/sbin/nologin
tss:x:106:113:TPM software stack,,:/var/lib/tpm:/bin/false
uiddd:x:107:116:/:/run/uiddd:/usr/sbin/nologin
systemd-oom:x:108:117:systemd Userspace OOM Killer,,,:/run/systemd:/usr/sbin/nologin
tcpdump:x:109:118:/:/nonexistent:/usr/sbin/nologin
avahi-autoipd:x:110:119:Avahi autoip daemon,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmuxd:x:111:46:usbmux daemon,,:/var/lib/usbmux:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,:/var/lib/misc:/usr/sbin/nologin
kernoops:x:113:65534:Kernel Oops Tracking Daemon,,:/usr/sbin/nologin
avahi:x:114:121:Avahi mDNS daemon,,:/run/avahi-daemon:/usr/sbin/nologin
cups-pk-helper:x:115:122:user for cups-pk-helper service,,:/home/cups-pk-helper:/usr/sbin/nologin
rtkit:x:116:123:RealtimeKit,,:/proc:/usr/sbin/nologin
whoopsie:x:117:124:/:/nonexistent:/bin/false
sssd:x:118:125:SSSD system user,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:119:29:Speech Dispatcher,,:/run/speech-dispatcher:/bin/false
fwupd-refresh:x:120:126:fwupd-refresh user,,:/run/systemd:/usr/sbin/nologin
nm-openvpn:x:121:127:NetworkManager OpenVPN,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
saned:x:122:129:/:/var/lib/saned:/usr/sbin/nologin
colord:x:123:130:colord colour management daemon,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:124:131:/:/var/lib/geoclue:/usr/sbin/nologin
pulse:x:125:132:PulseAudio daemon,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:126:65534:/:/run/gnome-initial-setup:/bin/false
hplip:x:127:7:HPLIP system user,,:/run/hplip:/bin/false
gdm:x:128:134:Gnome Display Manager:/var/lib/gdm3:/bin/false
rpkush:x:1000:1000:RUPALI KUSHWAHA,,:/home/rpkush:/bin/bash
rpkush@rpkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
rpkush@rpkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$ tail -n 5 /etc/passwd
pulse:x:125:132:PulseAudio daemon,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:126:65534:/:/run/gnome-initial-setup:/bin/false
hplip:x:127:7:HPLIP system user,,:/run/hplip:/bin/false
gdm:x:128:134:Gnome Display Manager:/var/lib/gdm3:/bin/false
rpkush:x:1000:1000:RUPALI KUSHWAHA,,:/home/rpkush:/bin/bash
rpkush@rpkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
nm-openvpn:x:121:127:NetworkManager OpenVPN,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
rpkush@rpkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$
```

Task 3 — Searching for Patterns

7. Find all lines containing the word root

Command

```
grep "root" /etc/passwd
```

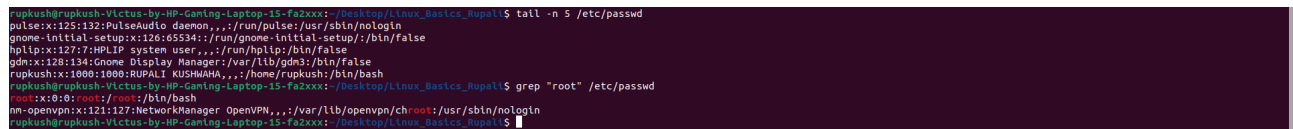
Explanation

The `grep` command is used to search for a particular text pattern inside a file. In this task, it searches `/etc/passwd` for lines containing the word `root`.

Example Output

```
root:x:0:0:root:/root:/bin/bash
nm-openvpn:x:121:127:NetworkManager
OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
```

Screenshot



```
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: /Desktop/Linux_Basics_Rupal$ tail -n 5 /etc/passwd
pulse:x:125:132:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:126:15534:,:/run/gnome-initial-setup:/bin/false
hplip:x:127:7:HP/HP system user,,,:/run/hplip:/bin/false
gdm:x:128:134:Gnome Display Manager:/var/lib/gdm3:/bin/false
rupkush:x:1000:1000:RUPALI KUSHWAHA,,,:/home/rupkush:/bin/bash
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: /Desktop/Linux_Basics_Rupal$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
nm-openvpn:x:121:127:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: /Desktop/Linux_Basics_Rupal$
```

Task 4 — Zipping and Unzipping

8. Compress the test_dir directory

Command

```
zip -r test_dir.zip test_dir
```

Explanation

The `zip` command is used to compress files and directories into a ZIP archive. The `-r` option allows the command to recursively include the contents of the directory.

The directory `test_dir` was compressed into:

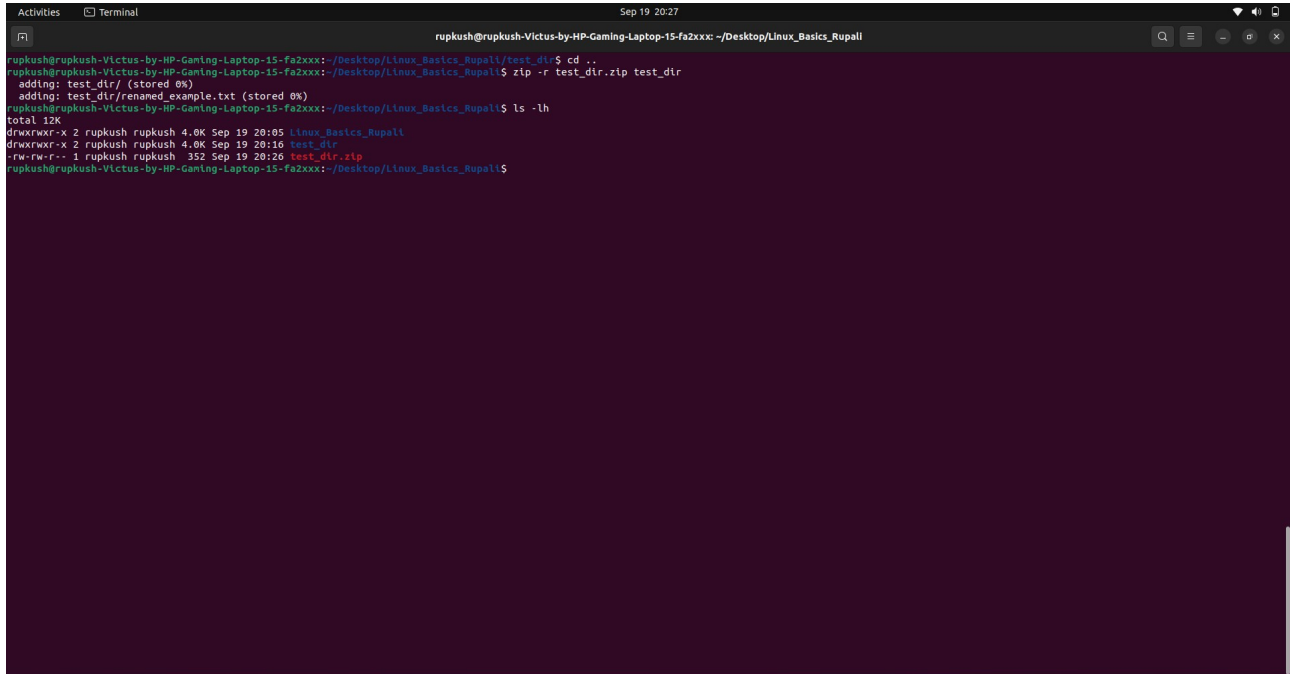
`test_dir.zip`

Verification

```
ls -lh
```

The ZIP file was successfully created.

Screenshot

A screenshot of a terminal window with a dark purple background. The window title is "Activities" and "Terminal". The terminal shows the following commands and output:

```
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali/test_dir$ cd ..
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$ zip -r test_dir.zip test_dir
adding: test_dir/ (stored 0%)
adding: test_dir/renamed_example.txt (stored 0%)
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$ ls -lh
total 12K
drwxrwxr-x 2 rupkush rupkush 4.0K Sep 19 20:05 Linux_Basics_Rupali
drwxrwxr-x 2 rupkush rupkush 4.0K Sep 19 20:16 test_dir
-rw-rw-r-- 1 rupkush rupkush 352 Sep 19 20:26 test_dir.zip
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$
```

9. Unzip test_dir.zip into unzipped_dir

Commands

```
mkdir unzipped_dir
unzip test_dir.zip -d unzipped_dir
```

Explanation

The `mkdir` command creates the destination directory `unzipped_dir`.

The `unzip` command extracts the contents of `test_dir.zip`. The `-d` option specifies the directory where the files should be extracted.

Verification

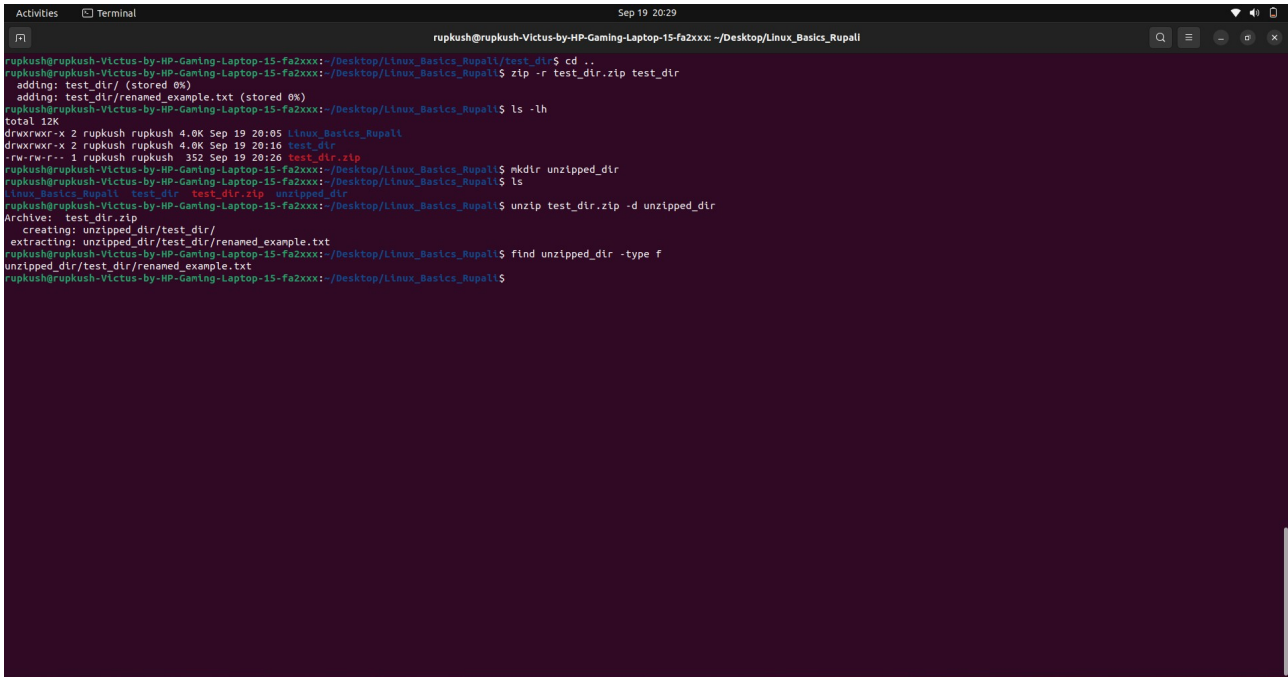
```
find unzipped_dir -type f
```

Output

```
unzipped_dir/test_dir/renamed_example.txt
```

This confirms that the ZIP archive was successfully extracted.

Screenshot



```
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx: ~/Desktop/Linux_Basics_Rupall
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx:~/Desktop/Linux_Basics_Rupall/test_dir$ cd ..
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx:~/Desktop/Linux_Basics_Rupall$ zip -r test_dir.zip test_dir
adding: test_dir/ (stored 0%)
adding: test_dir/renamed_example.txt (stored 0%)
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx:~/Desktop/Linux_Basics_Rupall$ ls -lh
total 12K
drwxrwxr-x 2 rupkush rupkush 4.0K Sep 19 20:05 Linux_Basics_Rupall
drwxrwxr-x 2 rupkush rupkush 4.0K Sep 19 20:16 test_dir
-rw-rw-r-- 1 rupkush rupkush 352 Sep 19 20:26 test_dir.zip
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx:~/Desktop/Linux_Basics_Rupall$ mkdir unzipped_dir
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx:~/Desktop/Linux_Basics_Rupall$ ls
Linux_Basics_Rupall  test_dir  test_dir.zip  unzipped_dir
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx:~/Desktop/Linux_Basics_Rupall$ unzip test_dir.zip -d unzipped_dir
Archive:  test_dir.zip
  creating: unzipped_dir/test_dir/
 extracting: unzipped_dir/test_dir/renamed_example.txt
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx:~/Desktop/Linux_Basics_Rupall$ find unzipped_dir -type f
unzipped_dir/test_dir/renamed_example.txt
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx:~/Desktop/Linux_Basics_Rupall$
```

Task 5 — Downloading Files

10. Download a file using `wget`

Command used

`wget`

`https://raw.githubusercontent.com/github/gitignore/main/Global/Linux.gitignore`

Explanation

`wget` is a command-line utility used to download files from the internet.

The provided example URL in the assignment returned a `404 Not Found` response because the requested file was not available at that URL. Therefore, a valid publicly available text file was used to demonstrate the `wget` command successfully.

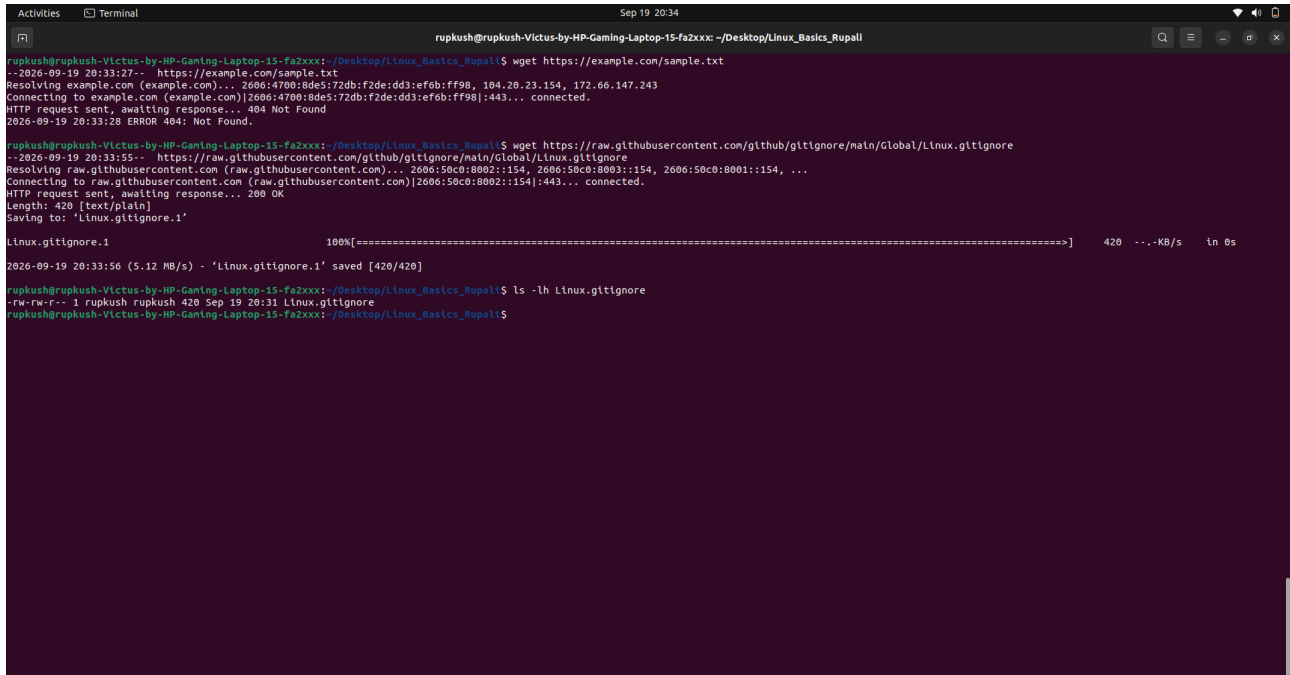
The downloaded file was saved as:

`Linux.gitignore`

Verification

`ls -lh Linux.gitignore`

Screenshot



```
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$ wget https://example.com/sample.txt
--2026-09-19 20:33:27-- https://example.com/sample.txt
Resolving example.com (example.com)... 2606:4700:8de5:72db:f2de:dd3:ef6b:ff98, 104.20.23.154, 172.66.147.243
Connecting to example.com (example.com)|2606:4700:8de5:72db:f2de:dd3:ef6b:ff98|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-09-19 20:33:28 ERROR 404: Not Found.

rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$ wget https://raw.githubusercontent.com/github/gitignore/main/Global/Linux.gitignore
--2026-09-19 20:33:55-- https://raw.githubusercontent.com/github/gitignore/main/Global/Linux.gitignore
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 2606:50c0:8002::154, 2606:50c0:8003::154, 2606:50c0:8001::154, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|2606:50c0:8002::154|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 420 [text/plain]
Saving to: 'Linux.gitignore.1'

Linux.gitignore.1                                100%[=====] 420 --.-KB/s  in 0s

2026-09-19 20:33:56 (5.12 MB/s) - 'Linux.gitignore.1' saved [420/420]

rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$ ls -lh Linux.gitignore
-rw-rw-r-- 1 rupkush rupkush 420 Sep 19 20:31 Linux.gitignore
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx:~/Desktop/Linux_Basics_Rupali$
```

Task 6 — Changing File Permissions

11. Create `secure.txt` and make it read-only for everyone

Create the file

```
touch secure.txt
```

Change permissions

```
chmod 444 secure.txt
```

Verify permissions

```
ls -l secure.txt
```

Expected permission format

```
-r--r--r--
```

Explanation

The `chmod` command is used to change file permissions.

- Owner: Read permission
- Group: Read permission

- Others: Read permission

No write or execute permissions are provided.

The permission representation:

`-r--r--r--`

means that the file is read-only for everyone.

Screenshot

```

rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$ touch secure.txt
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$ ls
Linux_Basics_Rupali Linux.gitignore.1 secure.txt test_dir test_dir.zip unzipped_dir
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$ ls -la
total 20
drwxrwxr-x 2 rupkush rupkush 4096 Sep 19 20:05 Linux_Basics_Rupali
-rw-rw-r-- 1 rupkush rupkush 420 Sep 19 20:33 Linux.gitignore.1
-rw-rw-r-- 1 rupkush rupkush 0 Sep 19 20:37 secure.txt
drwxrwxr-x 2 rupkush rupkush 4096 Sep 19 20:16 test_dir
-rw-rw-r-- 1 rupkush rupkush 352 Sep 19 20:26 test_dir.zip
drwxrwxr-x 3 rupkush rupkush 4096 Sep 19 20:28 unzipped_dir
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$ chmod -w secure.txt
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$ ls
Linux_Basics_Rupali Linux.gitignore.1 secure.txt test_dir test_dir.zip unzipped_dir
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$ ls -la
total 20
drwxrwxr-x 2 rupkush rupkush 4096 Sep 19 20:05 Linux_Basics_Rupali
-rw-rw-r-- 1 rupkush rupkush 420 Sep 19 20:33 Linux.gitignore.1
-r--r--r-- 1 rupkush rupkush 0 Sep 19 20:37 secure.txt
drwxrwxr-x 2 rupkush rupkush 4096 Sep 19 20:16 test_dir
-rw-rw-r-- 1 rupkush rupkush 352 Sep 19 20:26 test_dir.zip
drwxrwxr-x 3 rupkush rupkush 4096 Sep 19 20:28 unzipped_dir
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-fa2xxx: ~/Desktop/Linux_Basics_Rupali$

```

Task 7 — Working with Environment Variables

12. Create an environment variable named MY_VAR

Command

```
export MY_VAR="Hello, Linux!"
```

Explanation

The `export` command is used to create an environment variable that is available to the current shell and to processes started from that shell.

In this task, the variable `MY_VAR` was assigned the value:

Hello, Linux!

Verify using echo

```
echo $MY_VAR
```

Output

Hello, Linux!

Additional verification

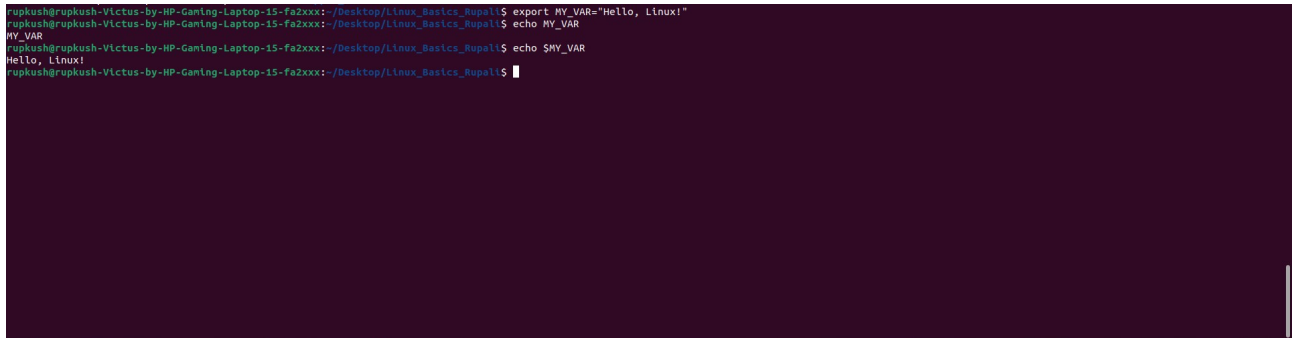
```
printenv MY_VAR
```

Output

Hello, Linux!

This confirms that the environment variable was successfully created.

Screenshot

A terminal window with a dark purple background. The prompt is 'rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx: ~/Desktop/Linux_Basics_Rupali\$'. The user enters 'export MY_VAR="Hello, Linux!"', followed by 'echo MY_VAR' which outputs 'MY_VAR'. Then they enter 'echo \$MY_VAR' which outputs 'Hello, Linux!'.

```
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx: ~/Desktop/Linux_Basics_Rupali$ export MY_VAR="Hello, Linux!"
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx: ~/Desktop/Linux_Basics_Rupali$ echo MY_VAR
MY_VAR
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx: ~/Desktop/Linux_Basics_Rupali$ echo $MY_VAR
Hello, Linux!
rupkush@rupkush-Victus-by-HP-Gaming-Laptop-15-Fa2xxx: ~/Desktop/Linux_Basics_Rupali$
```

Project Directory Structure

After completing all the tasks, the project directory contains the following files and directories:

```
Linux_Basics_Rupali/
├── Linux.gitignore
├── secure.txt
├── test_dir/
│   └── renamed_example.txt
├── test_dir.zip
├── unzipped_dir/
│   └── test_dir/
│       └── renamed_example.txt
```

Commands Practised

The following Linux commands were practised during this assignment:

Command	Purpose
<code>mkdir</code>	Create directories
<code>touch</code>	Create empty files
<code>mv</code>	Rename or move files
<code>ls</code>	List files and directories
<code>cat</code>	Display file contents
<code>head</code>	Display the beginning of a file
<code>tail</code>	Display the end of a file
<code>grep</code>	Search for patterns
<code>zip</code>	Compress files/directories
<code>unzip</code>	Extract ZIP archives
<code>wget</code>	Download files from the internet
<code>chmod</code>	Change file permissions
<code>export</code>	Set environment variables
<code>echo</code>	Display variable values
<code>printenv</code>	Display environment variables
<code>find</code>	Search for files/directories

Conclusion

This assignment provided practical experience with essential Linux command-line operations. I learned how to create and rename files and directories, view and search file contents, compress and extract directories, download files using `wget`, modify file permissions using `chmod`, and create environment variables using `export`.

These commands are fundamental Linux skills and are useful for system administration, software development, cloud computing, and DevOps workflows.

GitHub Repository

Repository Link:

<https://github.com/KUSHRUP/Linux-Basics-DevOps.git>

The GitHub repository contains the project files and documentation associated with this assignment.